

R Basic Course

This book was made with educational objetctive to improve R skill
in students and professional staff

Mardones, Mauricio & Zúñiga, María José.

2025-02-11

About Course_R

Description and educational material for introductory course on R, descriptive statistics, and visualizations with applications in fisheries.

This course has these sections;

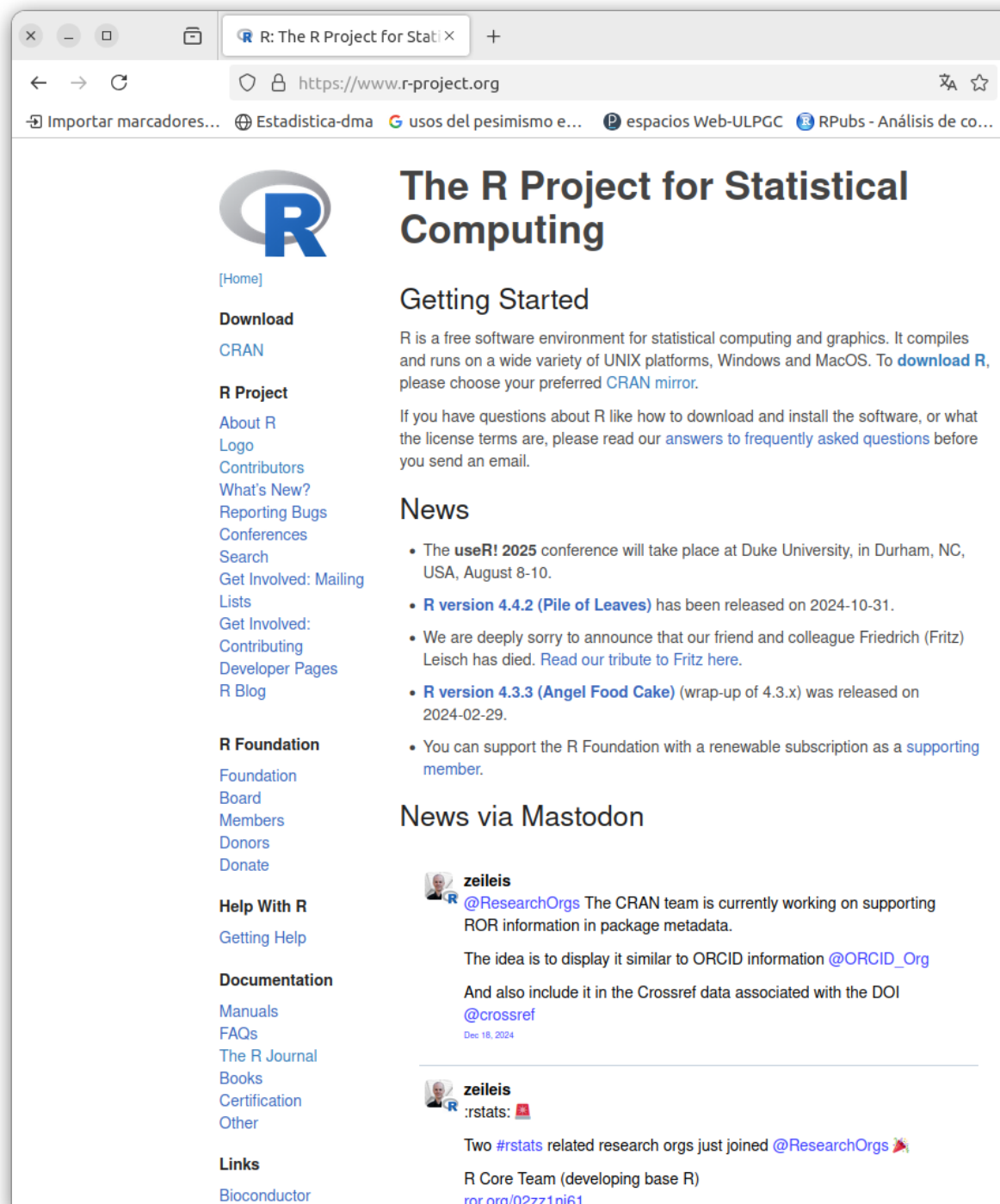
- 1. **useR**. Intro to R and basic features.
- 2. **Exploratory Data Analysis**. Basic concepts for data exploration. Essential statistical calculations.
- 3. **Visualization**. Graphics and Maps using modern packages
- 4. **Efficient code**. Organization strategies.
- 5. **Automatization and sharing** Report generation and result sharing.
- 6. **Management of Bibliographic** sources and additional materials.

All codes and used in this course can be founded in this repository with the structure to do this Ebook [Repo R Course](#)

useR. Intro to R and basic features

Welcome R user

- R is a programming language designed for:
 - Statistical computing
 - Data analysis
 - Visualization



The screenshot shows the official website of the R Project for Statistical Computing. The browser window has a single tab titled "R: The R Project for Stati...". The address bar shows "https://www.r-project.org". The page features the R logo, a navigation menu on the left, and main content sections for "Getting Started", "News", and "News via Mastodon".

The R Project for Statistical Computing

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Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred [CRAN mirror](#).

If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- The **useR! 2025** conference will take place at Duke University, in Durham, NC, USA, August 8-10.
- R version 4.4.2 (Pile of Leaves)** has been released on 2024-10-31.
- We are deeply sorry to announce that our friend and colleague Friedrich (Fritz) Leisch has died. [Read our tribute to Fritz here](#).
- R version 4.3.3 (Angel Food Cake)** (wrap-up of 4.3.x) was released on 2024-02-29.
- You can support the R Foundation with a renewable subscription as a [supporting member](#).

News via Mastodon

zeileis
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The idea is to display it similar to ORCID information [@ORCID_Org](#)

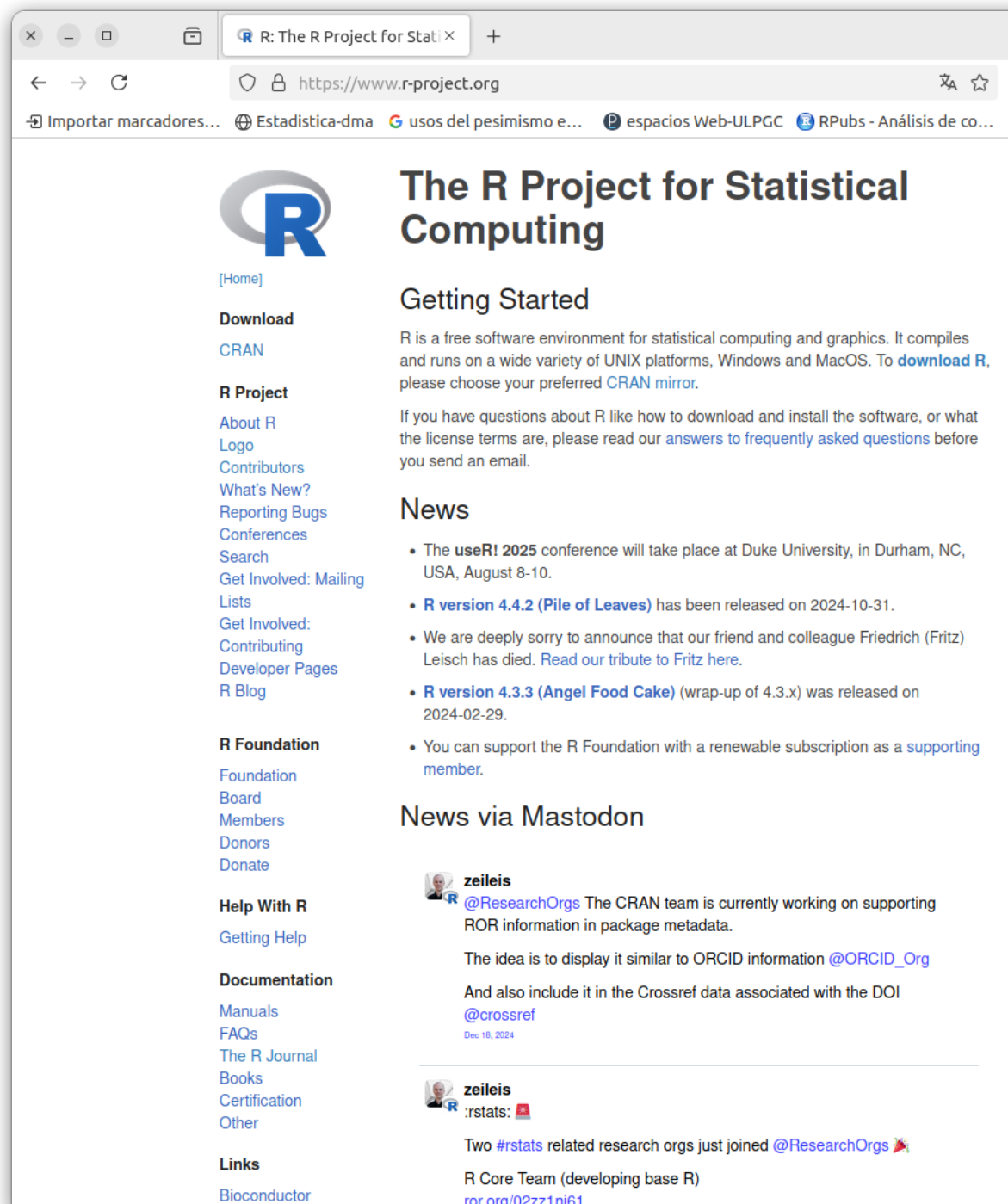
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R Core Team (developing base R)
[ror.org/02zz1nj61](#)

[Download R](#)



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Navigation Links (Left Sidebar):

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Main Content Area:

The R Project for Statistical Computing

Getting Started

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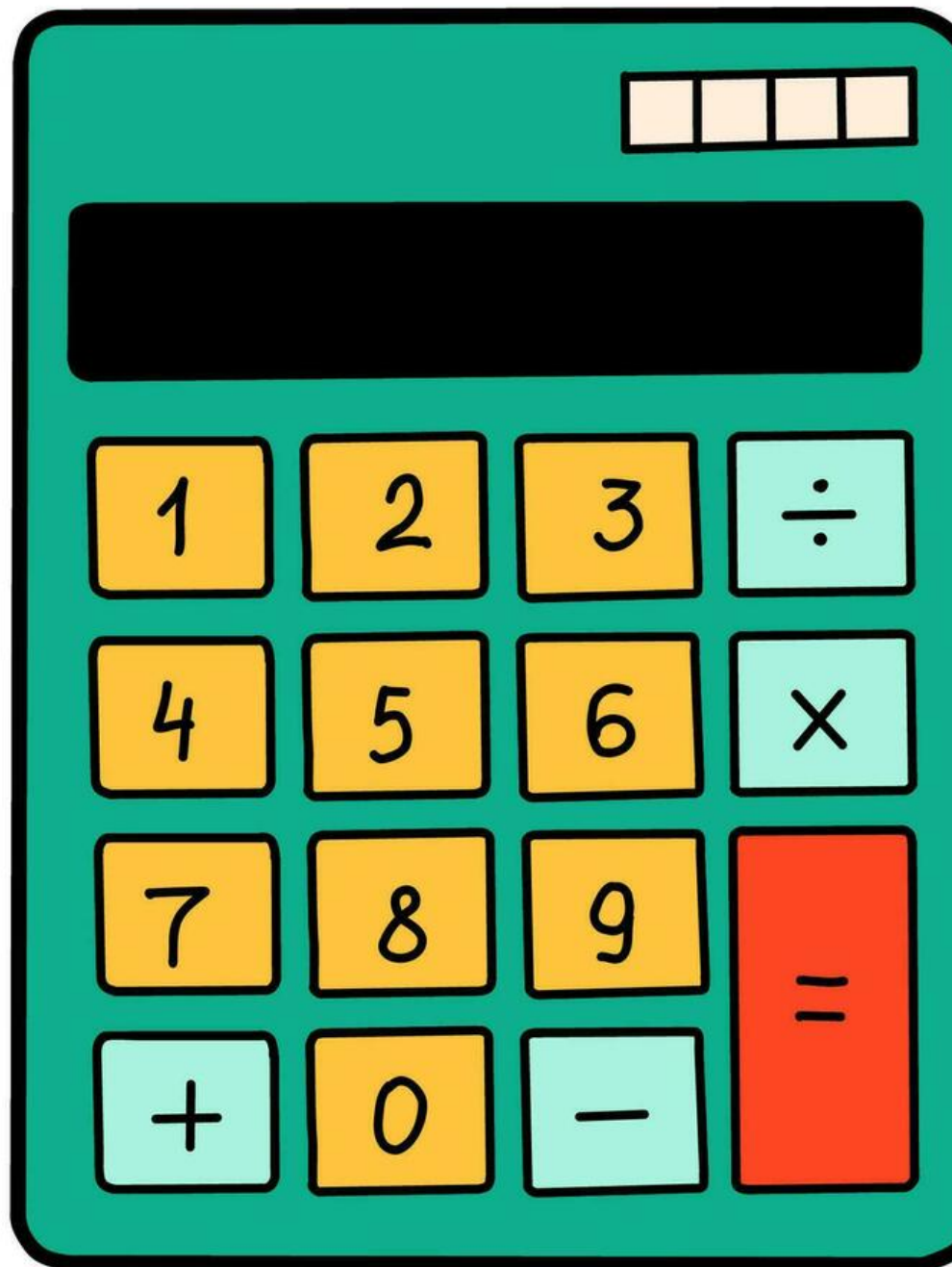
R Core Team (developing base R)
[ror.org/02zz1nj61](#)

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¿Why learn R?

- Open source and free
- big community
- ~ 13.000 packages
- Reproducible science

is just this



but...

Origin of R

Seminal paper on R! [here](#) (Fig. @ref(fig:semi)).

Getting Started

1. Install R from [CRAN](#)
2. Install RStudio (optional, but highly recommended)
3. Write your first R command:

```
print("Hello, world!")
```

```
## [1] "Hello, world!"
```

Basic Data Types

- **Numeric:** Numbers (e.g., 3.14, 42)
- **Character:** Text strings (e.g., "fish")
- **Logical:** TRUE or FALSE
- **Factor:** Categorical variables

Vectors

- R's most fundamental data structure:

```
numbers <- c(1, 2, 3, 4)
letters <- c("a", "b", "c")
```

- Operations are vectorized:

```
numbers * 2
```

```
## [1] 2 4 6 8
```

Data Frames

- Table-like structures:

```
df <- data.frame(Name = c("John", "Jane"), Age = c(25, 30))
```

- Inspect the structure:

```
head(df)
```

R: A Language for Data Analysis and Graphics

ROSS IHAKA and Robert GENTLEMAN

In this article we discuss our experience designing and implementing a statistical computing language. In developing this new language, we sought to combine what we felt were useful features from two existing computer languages. We feel that the new language provides advantages in the areas of portability, computational efficiency, memory management, and scoping.

Key Words: Computer language; Statistical computing.

1. INTRODUCTION

This article discusses some issues involved in the design and implementation of a computer language for statistical data analysis. Our experience with these issues occurred while developing such a language. The work has been heavily influenced by two existing languages—Becker, Chambers, and Wilks' S (1985) and Steel and Sussman's Scheme (1975). We felt that there were strong points in each of these languages and that it would be interesting to see if the strengths could be combined. The resulting language is very similar in appearance to S, but the underlying implementation and semantics are derived from Scheme. In fact, we implemented the language by first writing an interpreter for a Scheme subset and then progressively mutating it to resemble S.

We added S-like features in several stages. First, we altered the language parser so that the syntax would resemble that of S. This created a major change in the appearance of the language, but it should be emphasized that the change was entirely superficial; the underlying semantics remained those of Scheme. Next, we modified the data types of the language by removing the single scalar data type we had put into our Scheme and replacing it with the vector-based types of S. This was a much more substantive change and required major modifications to the interpreter. The final substantive change involved adding the S notion of *lazy arguments* for functions.

At this point we had enough of a framework in place to begin building a full statistical language. This process is ongoing, but we feel that we are well on the way to building a complete and useful piece of software. The development of the key portions of language

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and Interface Foundation of North America

Journal of Computational and Graphical Statistics, Volume 5, Number 3, Pages 299–314

Figure 1: Seminal Paper of R

```
##   Name Age
## 1 John  25
## 2 Jane  30
```

Visualization

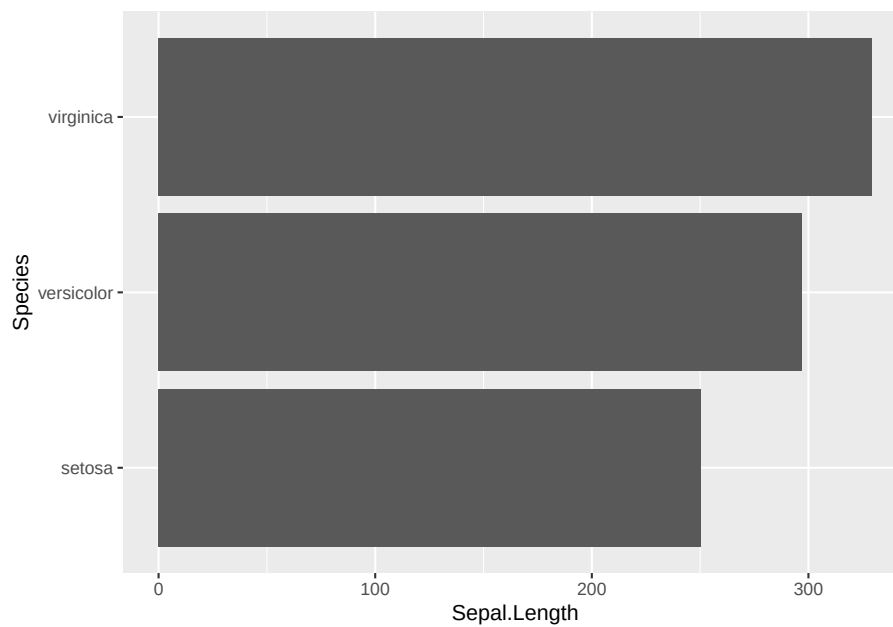
- Example base plot:

Data used

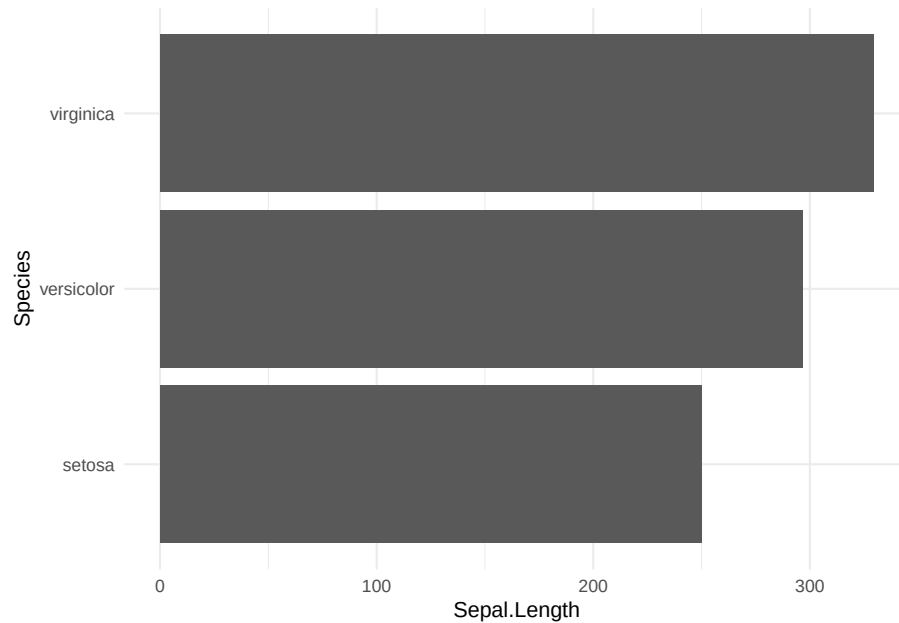
```
kbl(head(iris))
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

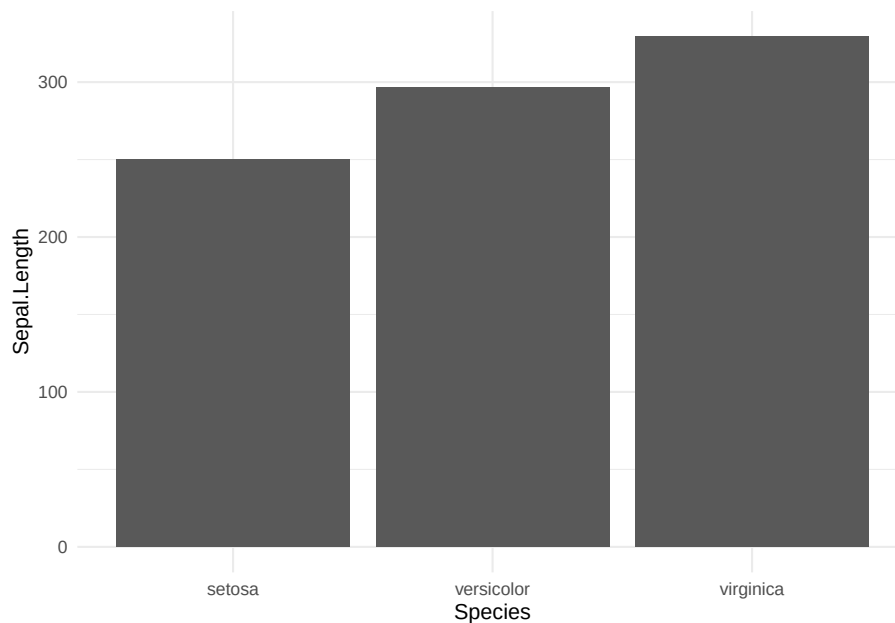
```
plottex <- ggplot(iris,
  aes(x = Sepal.Length,
      y = Species)) +
  geom_bar(stat = "identity")
plottex
```



```
plottex <- ggplot(iris,  
  aes(x = Sepal.Length,  
      y = Species)) +  
  geom_bar(stat = "identity")+  
  theme_minimal()  
plottex
```



```
plottex <- ggplot(iris,  
  aes(x = Sepal.Length,  
      y = Species)) +  
  geom_bar(stat = "identity")+  
  theme_minimal()+  
  coord_flip()  
plottex
```



Functions

- Encapsulate reusable logic:

```
square <- function(x) {  
  return(x^2)  
}  
square(4) # Output: 16
```

```
## [1] 16
```

Statistical Analysis

- Linear regression example:

```
fit <- lm(mpg ~ wt, data = mtcars)
```

```
summary(fit)
```

```
##  
## Call:  
## lm(formula = mpg ~ wt, data = mtcars)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max
```

```
## -4.5432 -2.3647 -0.1252  1.4096  6.8727
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  37.2851      1.8776  19.858 < 2e-16 ***
## wt          -5.3445      0.5591  -9.559 1.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 30 degrees of freedom
## Multiple R-squared:  0.7528, Adjusted R-squared:  0.7446
## F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

Conclusion

- R is powerful for:
 - Data analysis
 - Visualization
 - Statistics

Sources

Source in this [link](#)

[An introduction to R](#)

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a crucial first step in analyzing data. It provides a summary of the main characteristics of datasets and helps identify patterns, outliers, and relationships.

This section demonstrates an exploratory data analysis (EDA) of the **Palmer Penguins Dataset** using both base R and Tidyverse. It includes cleaning, descriptive statistics, and relationships between variables.

Load Necessary Packages

```
if (!require("palmerpenguins")) install.packages("palmerpenguins")

## Loading required package: palmerpenguins
if (!require("tidyverse")) install.packages("tidyverse")

library(palmerpenguins)
library(tidyverse)
```

Load and Clean Data

```
penguins <- palmerpenguins::penguins
penguins_clean <- drop_na(penguins)
```

Basic Data Overview

```
print(head(penguins_clean))

## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>          <dbl>         <dbl>         <int>         <int>
```

```
## 1 Adelie Torgersen      39.1      18.7      181      3750
## 2 Adelie Torgersen      39.5      17.4      186      3800
## 3 Adelie Torgersen      40.3      18      195      3250
## 4 Adelie Torgersen      36.7      19.3      193      3450
## 5 Adelie Torgersen      39.3      20.6      190      3650
## 6 Adelie Torgersen      38.9      17.8      181      3625
## # i 2 more variables: sex <fct>, year <int>
```

```
str(penguins_clean)
```

```
## tibble [333 x 8] (S3: tbl_df/tbl/data.frame)
## $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1
## $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3
## $ bill_length_mm : num [1:333] 39.1 39.5 40.3 36.7 39.3 38.9 39.2 41.1 38.6 34.6
## $ bill_depth_mm : num [1:333] 18.7 17.4 18 19.3 20.6 17.8 19.6 17.6 21.2 21.1
## $ flipper_length_mm: int [1:333] 181 186 195 193 190 181 195 182 191 198 ...
## $ body_mass_g    : int [1:333] 3750 3800 3250 3450 3650 3625 4675 3200 3800 4400
## $ sex            : Factor w/ 2 levels "female","male": 2 1 1 1 2 1 2 1 2 2 ...
## $ year           : int [1:333] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007
```

```
summary(select(penguins_clean, where(is.numeric)))
```

```
## bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
## Min. :32.10 Min. :13.10 Min. :172 Min. :2700
## 1st Qu.:39.50 1st Qu.:15.60 1st Qu.:190 1st Qu.:3550
## Median :44.50 Median :17.30 Median :197 Median :4050
## Mean :43.99 Mean :17.16 Mean :201 Mean :4207
## 3rd Qu.:48.60 3rd Qu.:18.70 3rd Qu.:213 3rd Qu.:4775
## Max. :59.60 Max. :21.50 Max. :231 Max. :6300
## year
## Min. :2007
## 1st Qu.:2007
## Median :2008
## Mean :2008
## 3rd Qu.:2009
## Max. :2009
```

Measures of Central Tendency

```
mean_flipper <- mean(penguins_clean$flipper_length_mm)
median_flipper <- median(penguins_clean$flipper_length_mm)
mean_flipper
```

```
## [1] 200.967
```

```
median_flipper
```

```
## [1] 197
```

Measures of Spread

```
range_body_mass <- range(penguins_clean$body_mass_g)
var_body_mass <- var(penguins_clean$body_mass_g)
sd_body_mass <- sd(penguins_clean$body_mass_g)

cat("\nRange of body mass:", range_body_mass)
```

```
##
## Range of body mass: 2700 6300

cat("\nVariance of body mass:", var_body_mass)
```

```
##
## Variance of body mass: 648372.5

cat("\nStandard deviation of body mass:", sd_body_mass)
```

```
##
## Standard deviation of body mass: 805.2158
```

Frequency Tables for Categorical Data

```
cat("\nFrequency of species:\n")
```

```
##
## Frequency of species:

print(table(penguins_clean$species))
```

```
##
##      Adelie Chinstrap   Gentoo
##      146         68     119
```

Relationships Between Variables

```
cov_bill <- cov(penguins_clean$bill_length_mm, penguins_clean$bill_depth_mm)
cor_bill <- cor(penguins_clean$bill_length_mm, penguins_clean$bill_depth_mm)
```

Outlier Detection

```
q1 <- quantile(penguins_clean$body_mass_g, 0.25)
q3 <- quantile(penguins_clean$body_mass_g, 0.75)
iqr_value <- q3 - q1
lower_bound <- q1 - 1.5 * iqr_value
upper_bound <- q3 + 1.5 * iqr_value

outliers <- penguins_clean %>%
  filter(body_mass_g < lower_bound | body_mass_g > upper_bound) %>%
  select(species, body_mass_g)

cat("\nOutliers in body mass:\n")

##
## Outliers in body mass:
print(outliers)

## # A tibble: 0 x 2
## #   i 2 variables: species <fct>, body_mass_g <int>
```

Comparison Between Groups

```
penguins_clean %>%
  filter(species %in% c("Adelie", "Gentoo")) %>%
  group_by(species) %>%
  summarise(
    mean_body_mass = mean(body_mass_g),
    sd_body_mass = sd(body_mass_g)
  )

## # A tibble: 2 x 3
##   species mean_body_mass sd_body_mass
##   <fct>         <dbl>         <dbl>
## 1 Adelie       3706.         459.
## 2 Gentoo      5092.         501.
```

Statistical Test: t-Test

```
penguins_subset <- penguins_clean %>%
  filter(species %in% c("Adelie", "Gentoo"))

t_test_result <- t.test(body_mass_g ~ species,
```

```

data = penguins_subset)

cat("\nT-Test Results:\n")

##
## T-Test Results:
print(t_test_result)

##
## Welch Two Sample t-test
##
## data: body_mass_g by species
## t = -23.254, df = 242.14, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Adelie and group Gentoo is not
## 95 percent confidence interval:
## -1503.702 -1268.843
## sample estimates:
## mean in group Adelie mean in group Gentoo
## 3706.164 5092.437

```

The null hypothesis (H0) assumes that the mean body mass of *Adelie* and *Gentoo* penguins is the same. If the p-value < 0.05, we reject H0, indicating a significant difference in the body mass of these species.

One-Way ANOVA

```

anova_body_mass <- aov(bill_length_mm ~ species, data = penguins_clean)
summary(anova_body_mass)

##              Df Sum Sq Mean Sq F value Pr(>F)
## species        2   7015    3508   397.3 <2e-16 ***
## Residuals     330    2914         9
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Simple Linear Regression

```

model <- lm(body_mass_g ~ flipper_length_mm, data = penguins_clean)
summary(model)

##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm, data = penguins_clean)
##

```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1057.33  -259.79   -12.24    242.97   1293.89
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -5872.09     310.29  -18.93  <2e-16 ***
## flipper_length_mm    50.15       1.54   32.56  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 393.3 on 331 degrees of freedom
## Multiple R-squared:  0.7621, Adjusted R-squared:  0.7614
## F-statistic: 1060 on 1 and 331 DF,  p-value: < 2.2e-16
```

Conclusion

This script demonstrates exploratory data analysis (EDA) using both Base R and Tidyverse, including cleaning, descriptive statistics, and relationships between variables. The statistical tests (t-test and ANOVA) provide insights into differences between groups, while regression analysis highlights potential predictive relationships.

Final Thoughts

- EDA is critical for understanding your data.
- Data handling ensures clean datasets.
- Data manipulation prepares data for in-depth analysis.
- Visualization helps reveal insights quickly.

References

[Basic R](#)

[Data Explorer](#)

[EMSE 4572 CleanData](#)

Visualization.

Graphics and Maps using modern packages

Graphics

Base R Visualizations

Load penguins dataset

```
if (!require("palmerpenguins")) install.packages("palmerpenguins")
library(palmerpenguins)
library(data.table)
```

```
##
## Attaching package: 'data.table'

## The following objects are masked from 'package:lubridate':
##
##     hour, isoweek, mday, minute, month, quarter, second, wday, week,
##     yday, year

## The following objects are masked from 'package:dplyr':
##
##     between, first, last

## The following object is masked from 'package:purrr':
##
##     transpose
data("penguins")
```

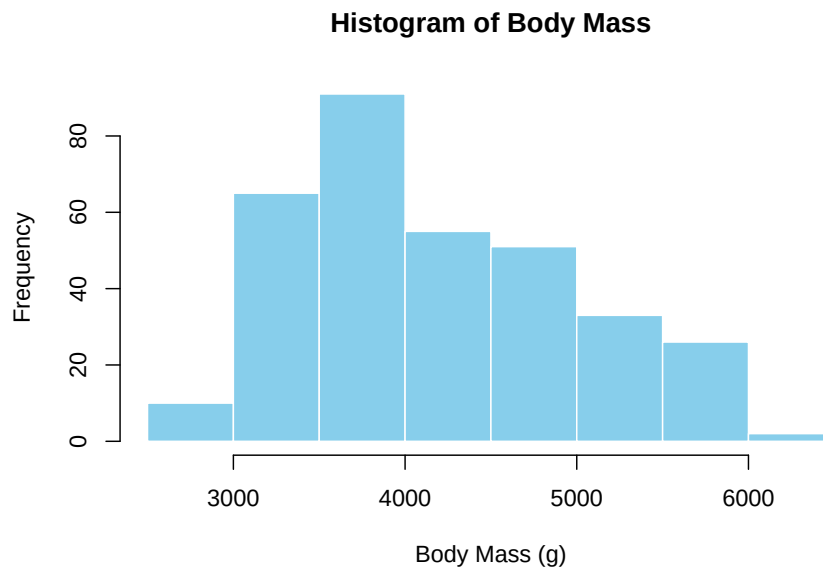
Load and Clean Data

```
penguins <- na.omit(palmerpenguins::penguins)
```

Visualization Techniques

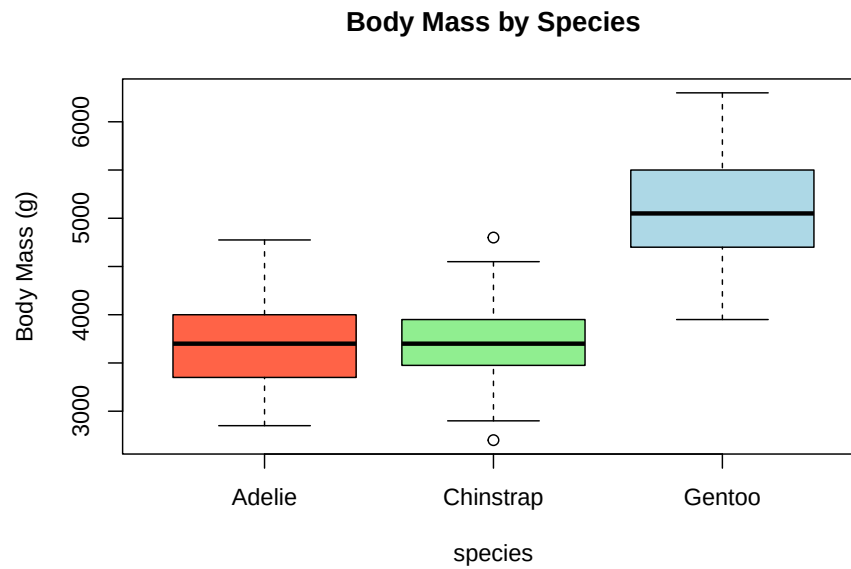
Histogram of body mass

```
hist(penguins$body_mass_g,  
     main = "Histogram of Body Mass",  
     xlab = "Body Mass (g)",  
     col = "skyblue",  
     border = "white")
```



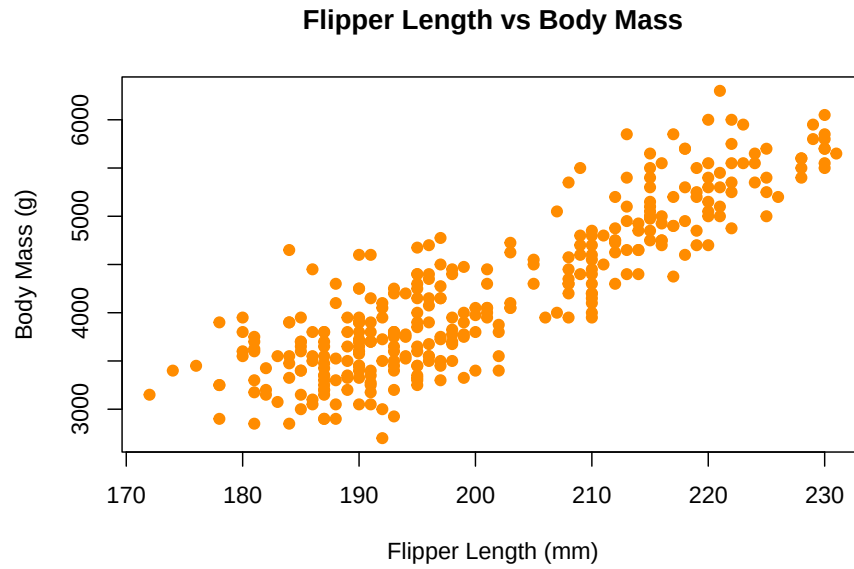
Boxplot of body mass by species

```
boxplot(body_mass_g ~ species, data = penguins,  
        main = "Body Mass by Species",  
        ylab = "Body Mass (g)",  
        col = c("tomato", "lightgreen", "lightblue"))
```

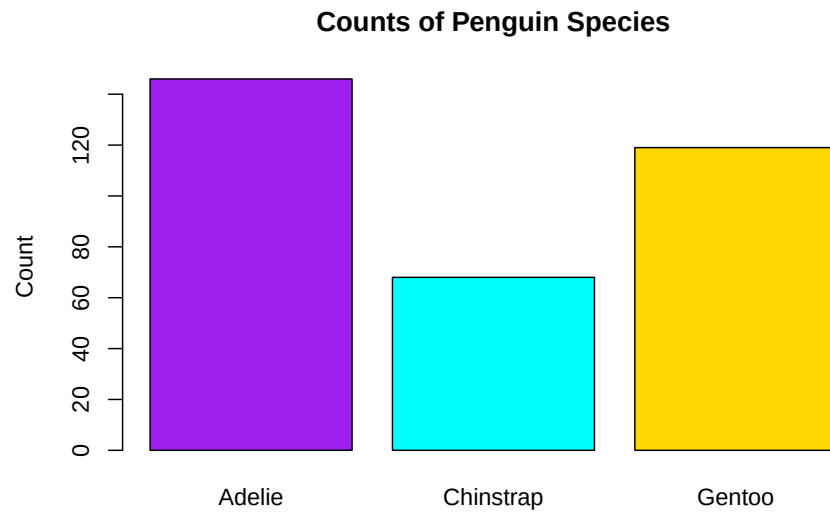
Scatter plot of flipper length vs body mass

```
plot(penguins$flipper_length_mm, penguins$body_mass_g,  
     main = "Flipper Length vs Body Mass",  
     xlab = "Flipper Length (mm)",  
     ylab = "Body Mass (g)",  
     pch = 19, col = "darkorange")
```



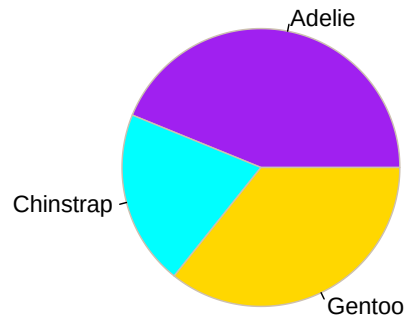
Bar plot of species counts

```
species_counts <- table(penguins$species)
barplot(species_counts,
        main = "Counts of Penguin Species",
        col = c("purple", "cyan", "gold"),
        ylab = "Count")
```



Pie chart of species distribution

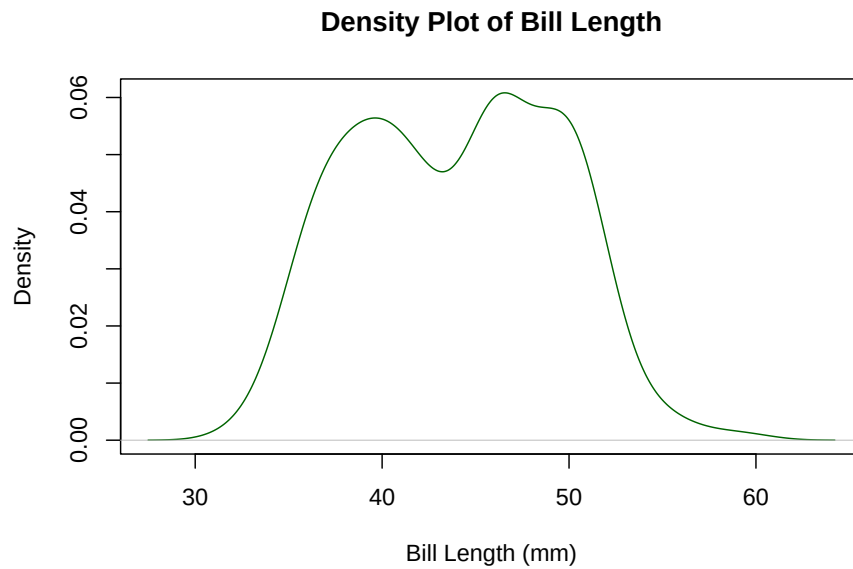
```
pie(species_counts,  
    main = "Species Distribution",  
    col = c("purple", "cyan", "gold"),  
    border = "antiquewhite3")
```

Species Distribution

```
# cl <- colors() N° colores
```

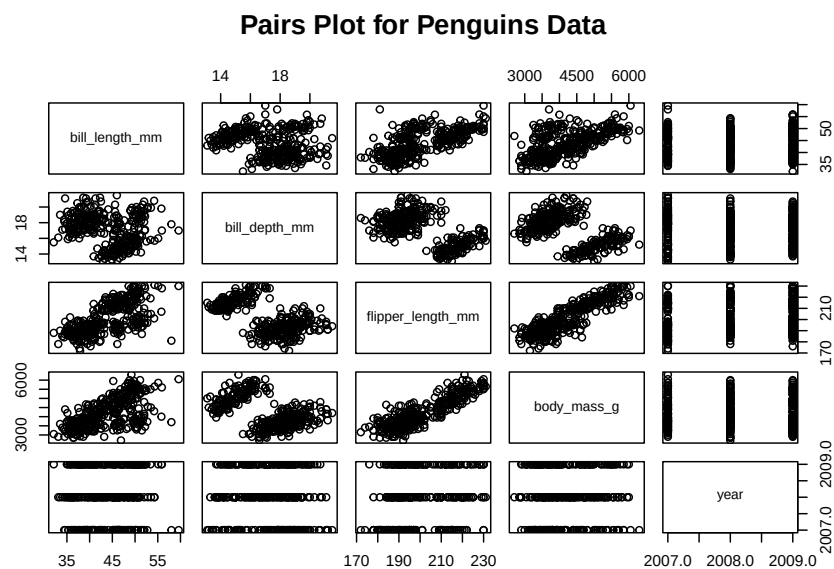
Density plot of bill length

```
plot(density(penguins$bill_length_mm),  
     main = "Density Plot of Bill Length",  
     xlab = "Bill Length (mm)",  
     col = "darkgreen")
```



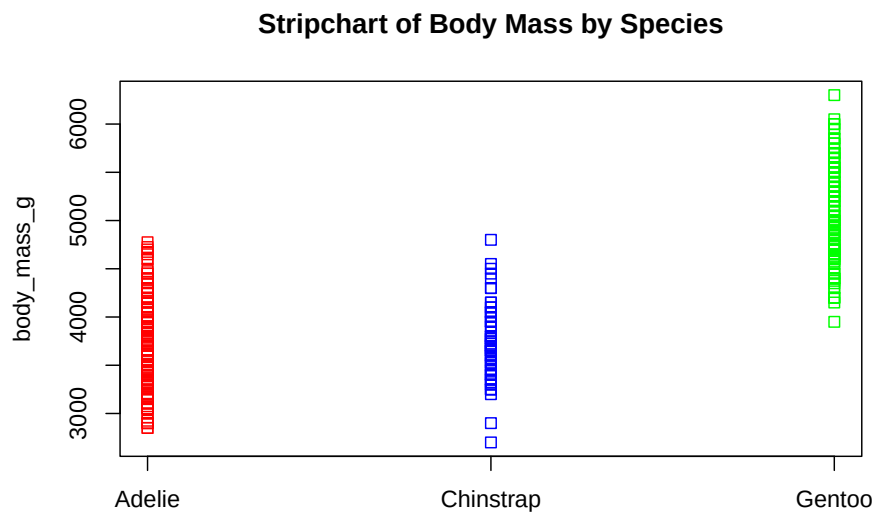
Pairs plot for numeric variables

```
pairs(penguins[, sapply(penguins, is.numeric)],
      main = "Pairs Plot for Penguins Data")
```



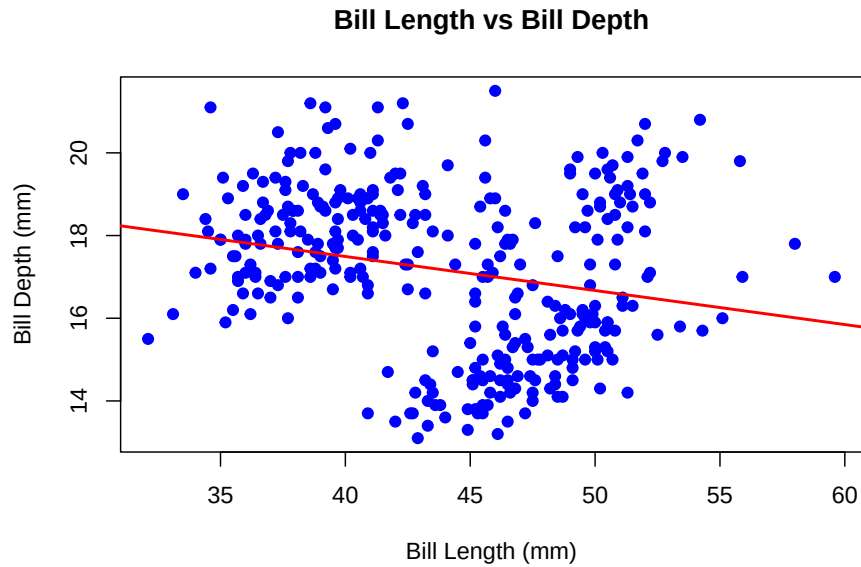
Stripchart of body mass by species

```
stripchart(body_mass_g ~ species, data = penguins,
           vertical = TRUE,
           main = "Stripchart of Body Mass by Species",
           col = c("red", "blue", "green"))
```



Scatter plot with regression line

```
plot(penguins$bill_length_mm, penguins$bill_depth_mm,
     main = "Bill Length vs Bill Depth",
     xlab = "Bill Length (mm)",
     ylab = "Bill Depth (mm)",
     pch = 19, col = "blue")
abline(lm(bill_depth_mm ~ bill_length_mm, data = penguins),
       col = "red", lwd = 2)
```



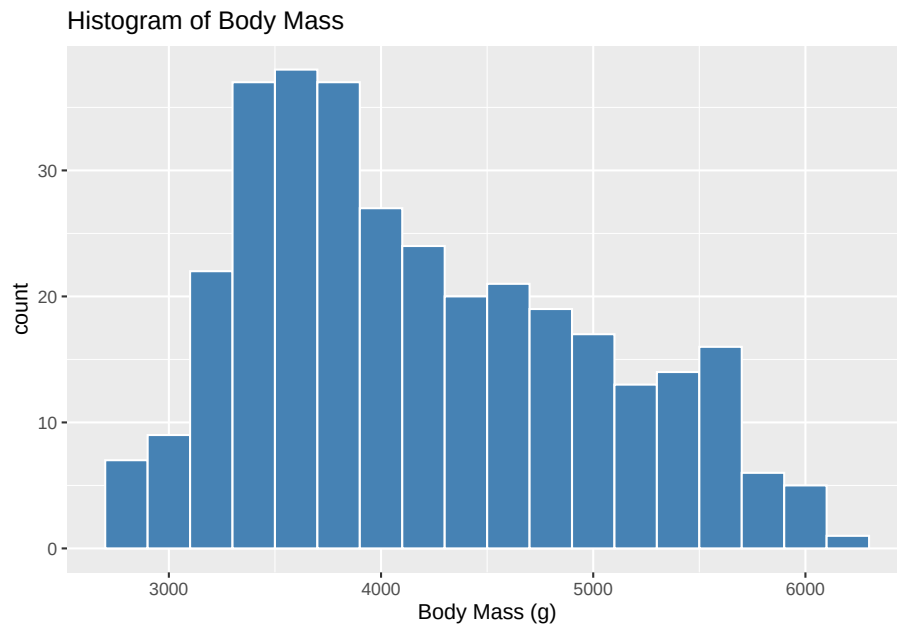
Tidyverse universe Visualizations

Load necessary package

```
if (!require("tidyverse")) install.packages("tidyverse")
library(tidyverse)
```

Histogram using ggplot2

```
ggplot(penguins, aes(x = body_mass_g)) +
  geom_histogram(binwidth = 200,
                 fill = "steelblue",
                 color = "white") +
  labs(title = "Histogram of Body Mass",
       x = "Body Mass (g)")
```

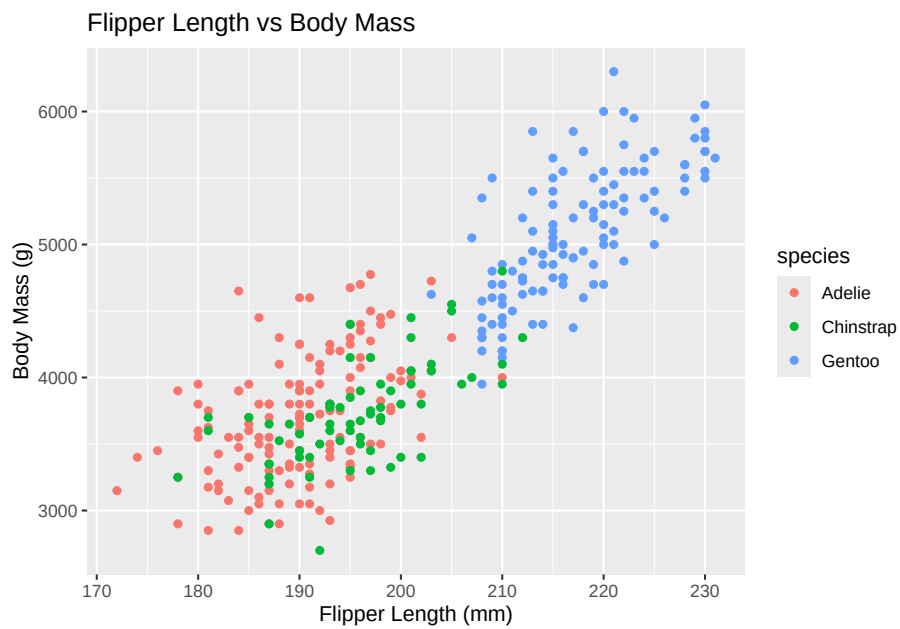


Boxplot of body mass by species

```
body <- ggplot(penguins, aes(x = species,  
                             y = body_mass_g,  
                             fill = species)) +  
  geom_boxplot() +  
  labs(title = "Body Mass by Species",  
       y = "Body Mass (g)")
```

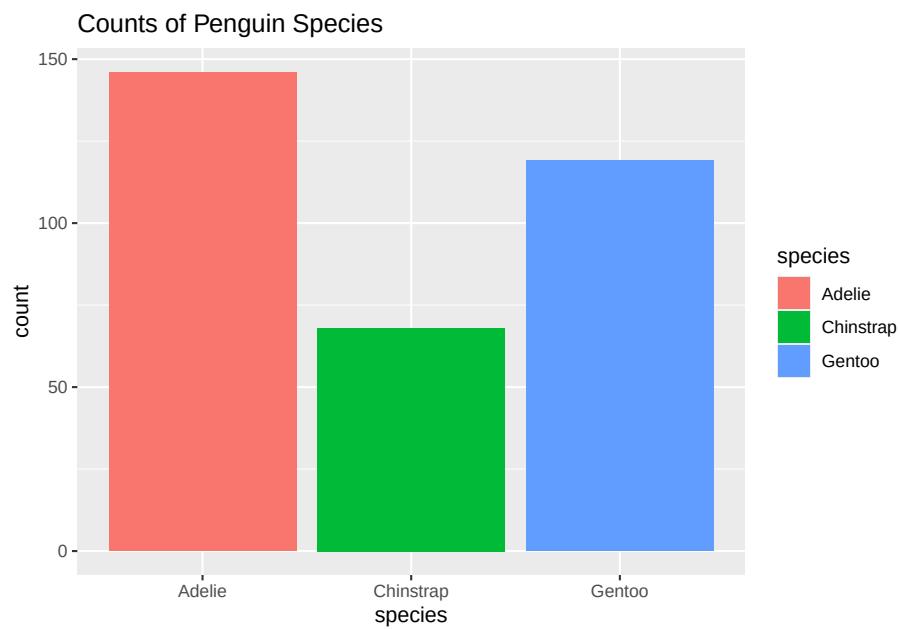
Scatter plot of flipper length vs body mass

```
ggplot(penguins, aes(x = flipper_length_mm,  
                     y = body_mass_g,  
                     color = species)) +  
  geom_point() +  
  labs(title = "Flipper Length vs Body Mass",  
       x = "Flipper Length (mm)",  
       y = "Body Mass (g)")
```

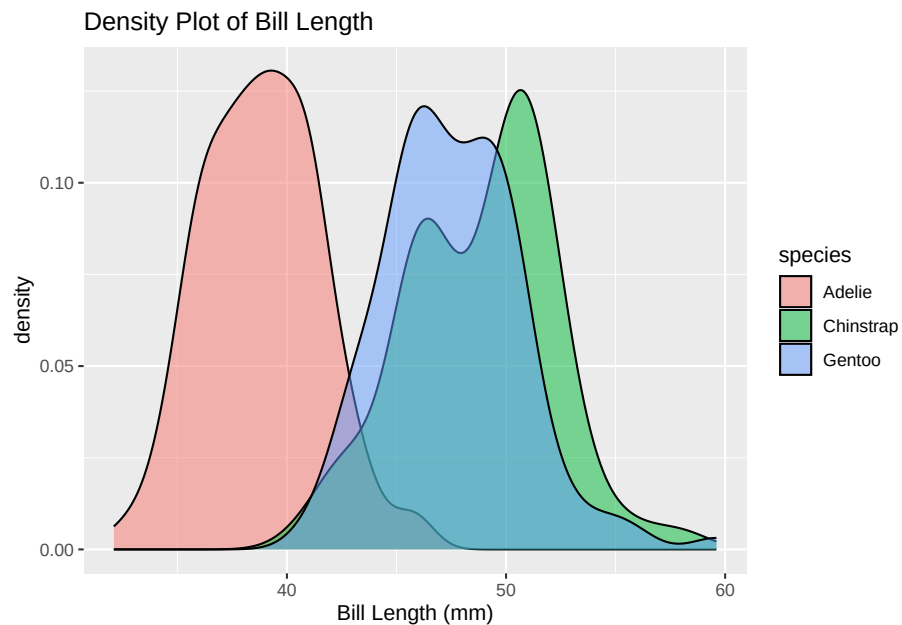
Bar plot of species counts

```
ggplot(penguins, aes(x = species, fill = species)) +  
  geom_bar() +  
  labs(title = "Counts of Penguin Species")
```



Density plot of bill length

```
ggplot(penguins,
       aes(x = bill_length_mm,
           fill = species)) +
  geom_density(alpha = 0.5) +
  labs(title = "Density Plot of Bill Length",
       x = "Bill Length (mm)")
```



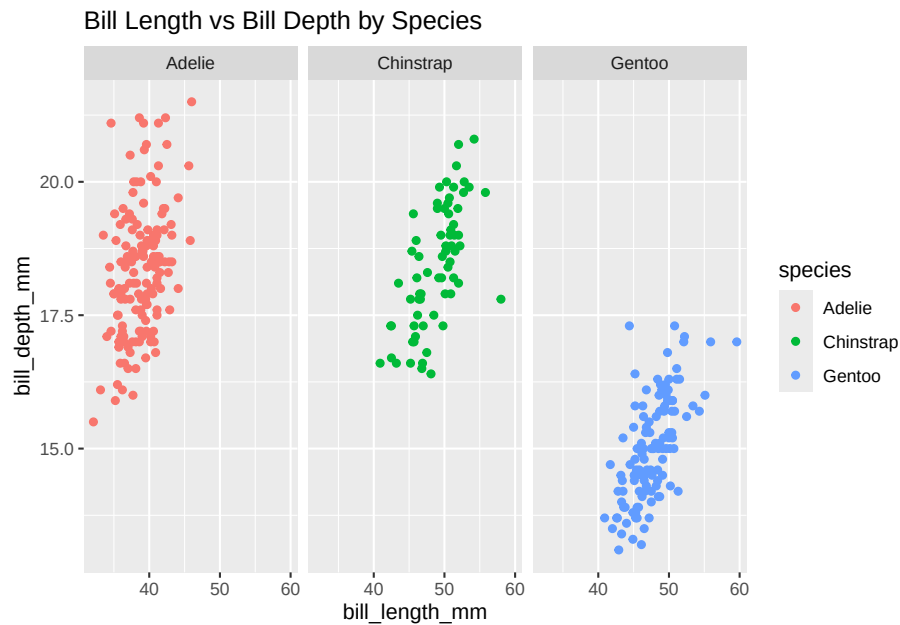
Violin plot of body mass by species

```
ggplot(penguins, aes(x = species,
                     y = body_mass_g,
                     fill = species)) +
  geom_violin() +
  labs(title = "Violin Plot of Body Mass by Species",
       y = "Body Mass (g)")
```



Faceted scatter plot

```
ggplot(penguins,  
  aes(x = bill_length_mm,  
      y = bill_depth_mm,  
      color = species)) +  
  geom_point() +  
  facet_wrap(~species) +  
  labs(title = "Bill Length vs Bill Depth by Species")
```



Correlations plot

Select numeric columns and compute correlation matrix

```
penguins_numeric <- penguins %>%
  select(where(is.numeric))
cor_matrix <- cor(penguins_numeric,
  use = "complete.obs")
```

Melt the correlation matrix for ggplot

```
cor_data <- data.table::melt(cor_matrix,
  varnames = c("Var1", "Var2"))
```

```
## Warning in melt.default(cor_matrix, varnames = c("Var1", "Var2")): The melt
## generic in data.table has been passed a matrix and will attempt to redirect to
## the relevant reshape2 method; please note that reshape2 is superseded and is no
## longer actively developed, and this redirection is now deprecated. To continue
## using melt methods from reshape2 while both libraries are attached, e.g.
## melt.list, you can prepend the namespace, i.e. reshape2::melt(cor_matrix). In
## the next version, this warning will become an error.
```

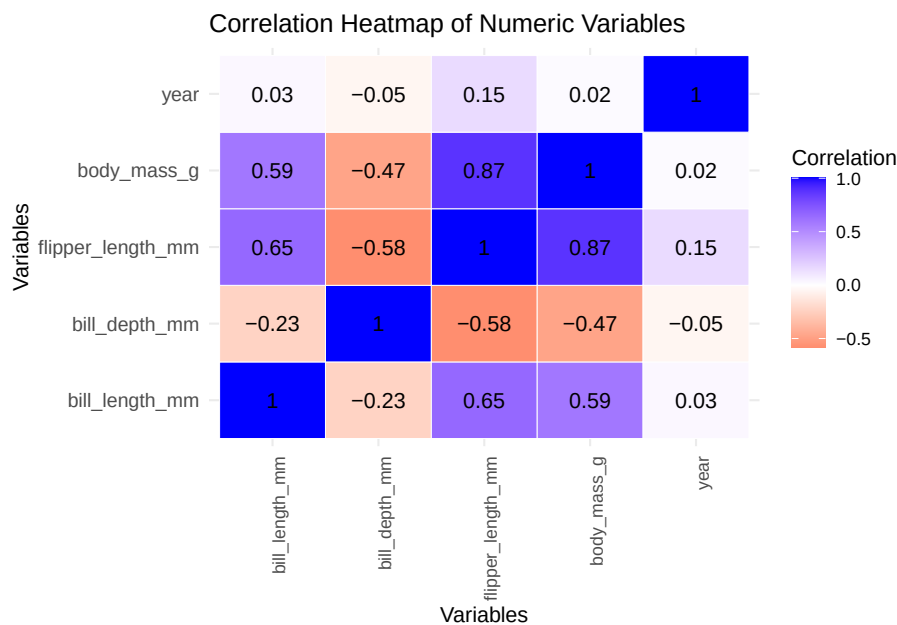
Create the heatmap plot

```
ggplot(cor_data, aes(x = Var1,
  y = Var2,
  fill = value)) +
  geom_tile(color = "white") +
```

```

scale_fill_gradient2(low = "red", mid = "white",
                    high = "blue",
                    midpoint = 0) +
geom_text(aes(label = round(value, 2)), color = "black", size = 4) +
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 90, hjust = 1),
  axis.text.y = element_text(size = 10)
) +
labs(
  title = "Correlation Heatmap of Numeric Variables",
  x = "Variables",
  y = "Variables",
  fill = "Correlation"
)

```



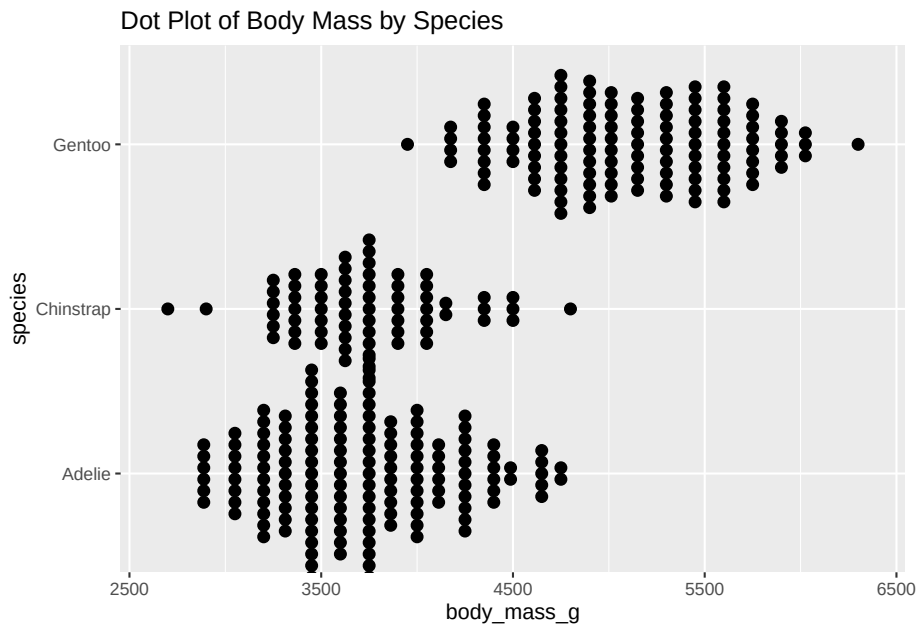
Dot plot

```

ggplot(penguins,
  aes(x = body_mass_g,
      y = species)) +
geom_dotplot(binaxis = "x",
  stackdir = "center",
  dotsize = 0.5) +
labs(title = "Dot Plot of Body Mass by Species")

```

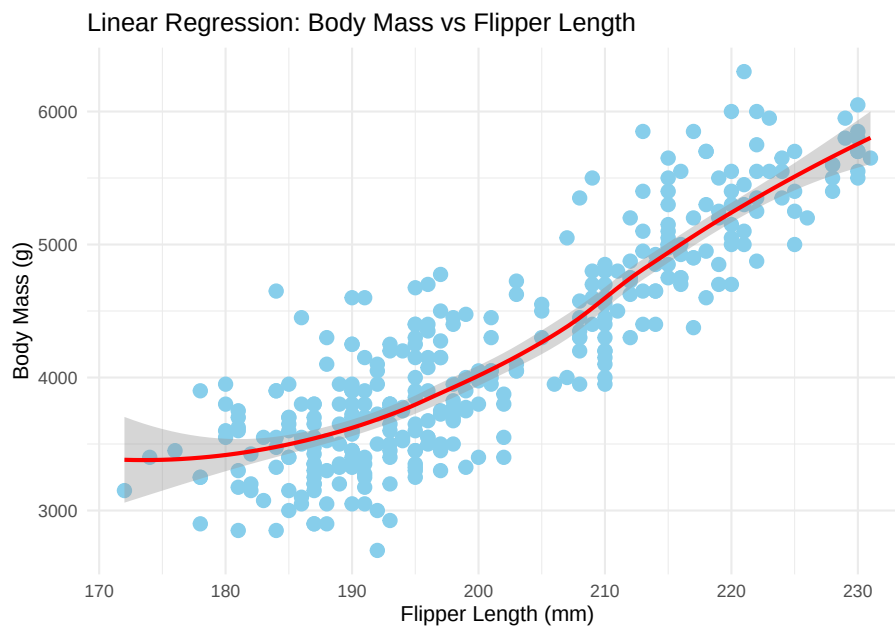
```
## Bin width defaults to 1/30 of the range of the data. Pick better value with
## 'binwidth'.
```



Plot regression with tidyverse

```
penguins %>%
  ggplot(aes(x = flipper_length_mm, y = body_mass_g)) +
  geom_point(color = "skyblue", size = 3) +
  geom_smooth(method = "loess",
              color = "red", se = TRUE) +
  labs(title = "Linear Regression: Body Mass vs Flipper Length",
       x = "Flipper Length (mm)",
       y = "Body Mass (g)") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Multivariate Analysis Example (PCA)

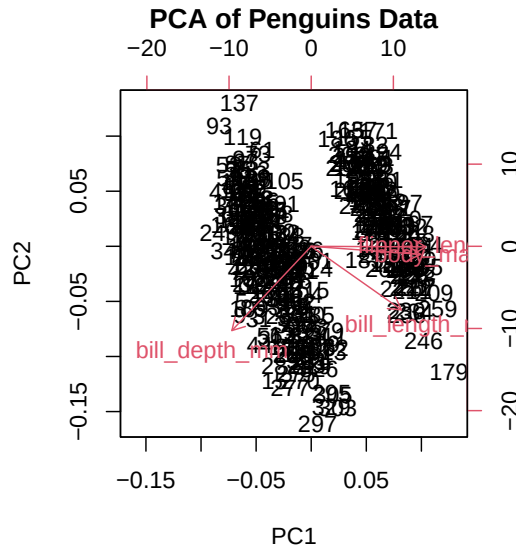
Base R PCA

```
pca_data <- penguins %>%
  select(c("bill_length_mm",
           "bill_depth_mm",
           "flipper_length_mm",
           "body_mass_g"))
pca_result <- prcomp(na.omit(pca_data),
  scale. = TRUE)
summary(pca_result)
```

```
## Importance of components:
##              PC1      PC2      PC3      PC4
## Standard deviation  1.6569 0.8821 0.60716 0.32846
## Proportion of Variance 0.6863 0.1945 0.09216 0.02697
## Cumulative Proportion 0.6863 0.8809 0.97303 1.00000
```

PCA Plot in Base R

```
biplot(pca_result, main = "PCA of Penguins Data")
```



Clustering with k-means

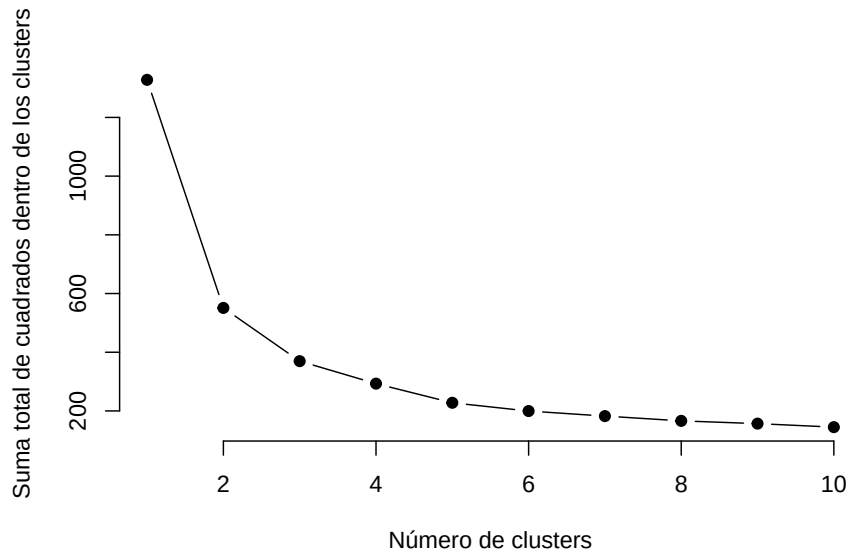
```
library(cluster)

# Seleccionar solo las columnas numéricas necesarias (removiendo NAs)
data <- na.omit(penguins[, c("bill_length_mm", "bill_depth_mm",
                             "flipper_length_mm", "body_mass_g")])

# Escalado de los datos
scaled_data <- scale(data)

# Selección del número óptimo de clústeres (Método Elbow)
wss <- sapply(1:10, function(k) {
  kmeans(scaled_data, centers = k, nstart = 25)$tot.withinss
})

# Gráfico del codo
plot(1:10, wss, type = "b", pch = 19, frame = FALSE,
     xlab = "Número de clusters",
     ylab = "Suma total de cuadrados dentro de los clusters")
```

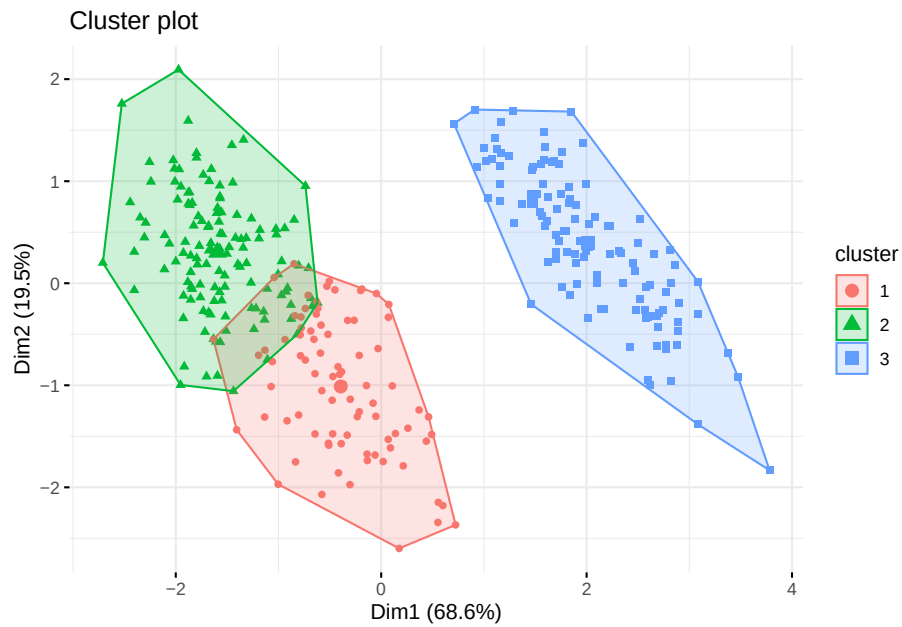



```
# Aplicación del algoritmo k-means
kmeans_result <- kmeans(scaled_data, centers = 3,
                        nstart = 25)
data$cluster <- as.factor(kmeans_result$cluster)

# Visualización de clústeres
library(factoextra)
```

Welcome! Want to learn more? See two factoextra-related books at <https://goo.gl/ve3WBa>

```
# Visualización en un espacio PCA
fviz_cluster(kmeans_result,
             data = scaled_data,
             geom = "point", ellipse.type = "convex",
             ggtheme = theme_minimal())
```



Simple maps

Instalar paquetes necesarios si no están instalados.

```
library(sf)
```

```
## Linking to GEOS 3.11.0, GDAL 3.5.3, PROJ 9.1.0; sf_use_s2() is TRUE
```

```
library(ggplot2)
library(dplyr)
library(ggspatial)
library(rnaturalearth)
library(rnaturalearthdata)
```

```
##
## Attaching package: 'rnaturalearthdata'

## The following object is masked from 'package:rnaturalearth':
##
## countries110
```

```
library(terra)
```

```
## terra 1.7.78
```

```
##
## Attaching package: 'terra'
```

```
## The following object is masked from 'package:data.table':  
##  
##      shift  
  
## The following object is masked from 'package:tidyr':  
##  
##      extract
```

```
library(lubridate)
```

Chile example:

```
chile <- ne_states(country = "Chile", returnclass = "sf")
```

Puerto Montt example

```
region_puerto_montt <- chile %>%  
  filter(name_en == "Los Lagos")
```

random coordinates with variable

```
lat_center <- -41.469  
lon_center <- -72.942  
  
#set.seed(42) # Semilla para reproducibilidad  
n_points <- 50  
points_data <- data.frame(  
  lon = runif(n_points, lon_center - 0.8, lon_center + 0.8),  
  lat = runif(n_points, lat_center - 0.8, lat_center + 0.8),  
  respiration_rate = rnorm(n_points, mean = 10, sd = 2)  
)
```

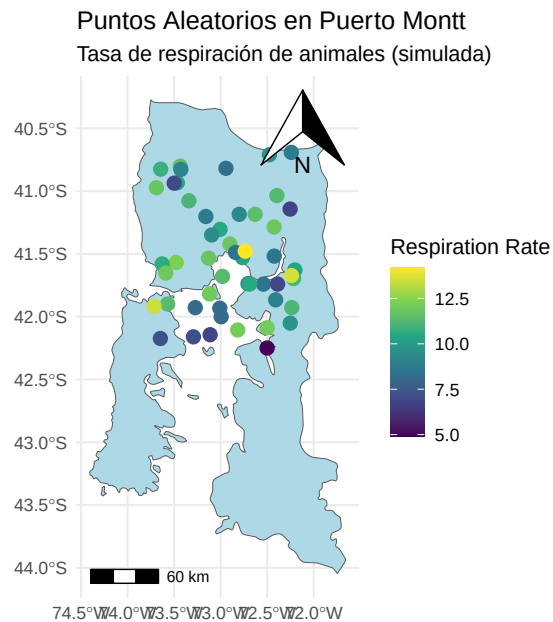
Convert sf object

```
points_sf <- st_as_sf(points_data, coords = c("lon", "lat"), crs = 4326)
```

Test map with attributes

```
plot <- ggplot(data = region_puerto_montt) +  
  geom_sf(fill = "lightblue") +  
  geom_sf(data = points_sf, aes(color = respiration_rate), size = 3) +  
  scale_color_viridis_c() +  
  annotation_scale(location = "bl") +  
  annotation_north_arrow(location = "tr", which_north = "true") +  
  labs(  
    title = "Puntos Aleatorios en Puerto Montt",  
    subtitle = "Tasa de respiración de animales (simulada)",  
    color = "Respiration Rate"  
  ) +  
  theme_minimal()
```

```
print(plot)
```



Convertir a objeto espacial `sf`

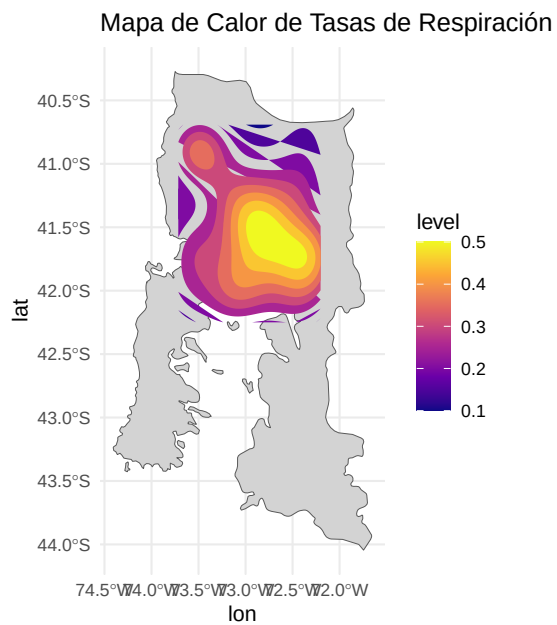
```
points_sf <- st_as_sf(points_data, coords = c("lon", "lat"), crs = 4326)
```

Other maps

Heat map

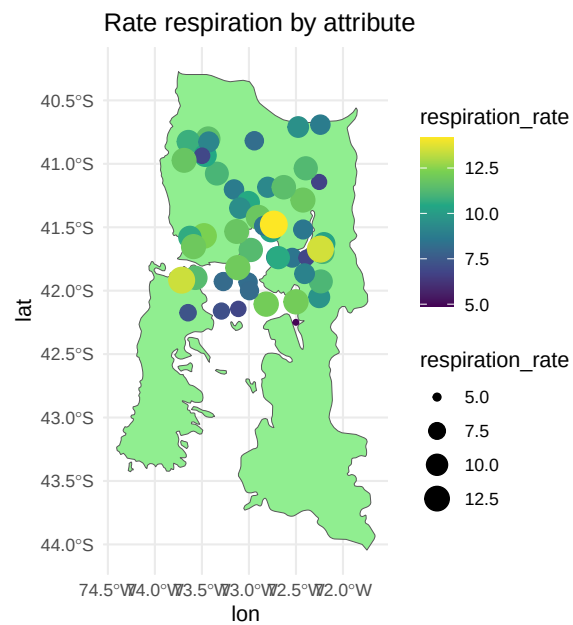
```
ggplot(data = region_puerto_montt) +  
  geom_sf(fill = "lightgrey") +  
  stat_density_2d(data = points_data,  
    aes(x = lon, y = lat,  
      fill = ..level..),  
    geom = "polygon") +  
  scale_fill_viridis_c(option="C") +  
  labs(title = "Mapa de Calor de Tasas de Respiración") +  
  theme_minimal()
```

```
## Warning: The dot-dot notation ('..level..') was deprecated in ggplot2 3.4.0.  
## i Please use 'after_stat(level)' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```



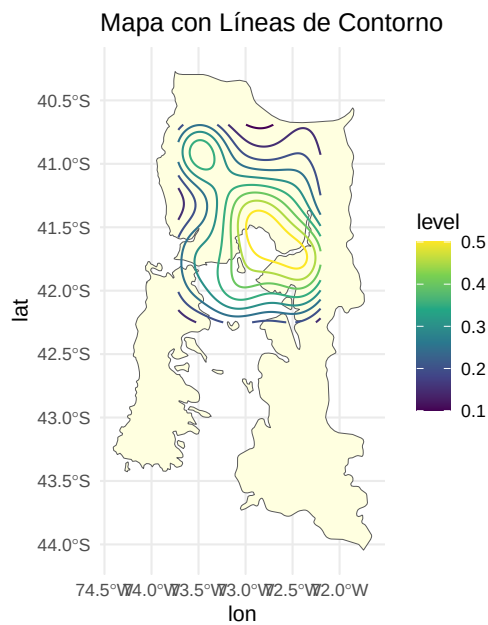
Points by attribute

```
ggplot(data = region_puerto_montt) +
  geom_sf(fill = "lightgreen") +
  geom_point(data = points_data,
    aes(x = lon, y = lat,
      size = respiration_rate,
      color = respiration_rate)) +
  scale_color_viridis_c() +
  labs(title = "Rate respiration by attribute") +
  theme_minimal()
```



Mapa con líneas de contorno

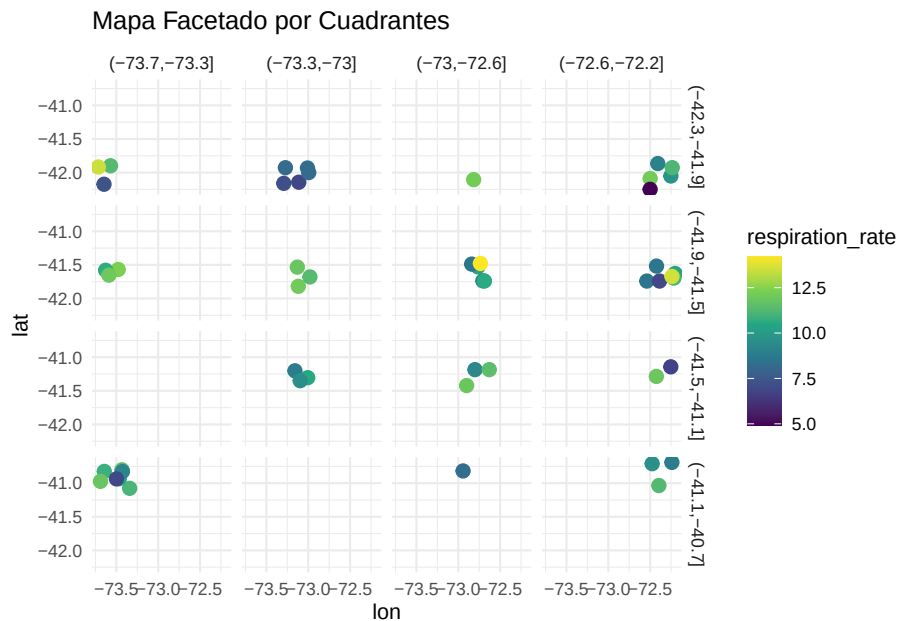
```
ggplot(data = region_puerto_montt) +
  geom_sf(fill = "lightyellow") +
  stat_density_2d(data = points_data, aes(x = lon, y = lat, color = ..level..), geom =
  scale_color_viridis_c() +
  labs(title = "Mapa con Líneas de Contorno") +
  theme_minimal()
```



Mapa por cuadrantes

```
points_data <- points_data %>%
  mutate(
    lat_group = cut(lat, breaks = 4),
    lon_group = cut(lon, breaks = 4)
  )

ggplot(points_data, aes(x = lon, y = lat, color = respiration_rate)) +
  geom_point(size = 3) +
  facet_grid(lat_group ~ lon_group) +
  scale_color_viridis_c() +
  labs(title = "Mapa Facetado por Cuadrantes") +
  theme_minimal()
```



```
# Cargar shapefile en R
# ruta_shapefile <- "ruta/del/archivo.shp"
# shapefile_data <- st_read(ruta_shapefile)
# plot(st_geometry(shapefile_data))

# Cargar archivo Excel con datos georreferenciados
# ruta_excel <- "ruta/del/archivo.xlsx"
# datos_excel <- read_excel(ruta_excel)
# sf_datos <- st_as_sf(datos_excel, coords = c("lon", "lat"), crs = 4326)
# ggplot() +
#   geom_sf(data = region_puerto_montt, fill = "lightgrey") +
#   geom_sf(data = sf_datos, color = "red", size = 3) +
#   theme_minimal()
```

The **sf** package (Simple Features) by Edzer Pebesma is a crucial tool for spatial data analysis in R. It provides a modern, efficient, and standardized way to handle vector spatial data using the Simple Features specification defined by the Open Geospatial Consortium (OGC). Here's why it's significant and why it has become essential compared to older, deprecated libraries:

Key Advantages of **sf**:

1. Simple Features Standard:

- Adheres to the OGC Simple Features standard, which ensures compatibility with modern GIS systems (e.g., QGIS, PostGIS).

2. Enhanced Performance:

- Operates on top of the GEOS, GDAL, and PROJ libraries, offering high-performance spatial operations.
 - More memory-efficient than older alternatives like `sp`.
3. **Easy Integration with the Tidyverse:**
 - `sf` objects (data frames with a geometry column) integrate seamlessly with `dplyr`, `ggplot2`, and other tidyverse packages.
 4. **Better Handling of Coordinate Systems:**
 - Simplifies the manipulation and transformation of coordinate reference systems (CRS).
 - Easily handles projections via the `st_crs()` function.
 5. **Deprecated Libraries:**
 - Older packages like `sp` and `rgdal` have been phased out due to their complex S4 structure and lack of maintenance.
 - `sf` uses a simpler and more flexible approach with class `sfc` (simple feature column), making operations straightforward.

Useful Functions

- `st_read()`: Import spatial data (shapefiles, geoJSON, etc.).
 - `st_transform()`: Change CRS.
 - `st_intersection()`, `st_union()`: Spatial operations.
 - `ggplot2::geom_sf()`: Plot spatial data directly.
-

Additional Resources

- [Official `sf` Documentation](#)
- [CRAN Page for `sf`](#)
- [Simple Features Standard by OGC](#)
- [R-spatial Blog](#) for spatial data tutorials

Efficient Code

Equations

Here is an equation.

$$f(k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ (\#eq : binom)} \quad (1)$$

You may refer to using `\@ref{eq:binom}`, like see Equation `@ref{eq:binom}`.

Read more here <https://bookdown.org/yihui/bookdown/markdown-extensions-by-bookdown.html>.

Callout blocks

The R Markdown Cookbook provides more help on how to use custom blocks to design your own callouts: <https://bookdown.org/yihui/rmarkdown-cookbook/custom-blocks.html>

Reporting and Sharing

Publishing

HTML books can be published online, see: <https://bookdown.org/yihui/bookdown/publishing.html>

404 pages

By default, users will be directed to a 404 page if they try to access a webpage that cannot be found. If you'd like to customize your 404 page instead of using the default, you may add either a `_404.Rmd` or `_404.md` file to your project root and use code and/or Markdown syntax.

Metadata for sharing

Bookdown HTML books will provide HTML metadata for social sharing on platforms like Twitter, Facebook, and LinkedIn, using information you provide in the `index.Rmd` YAML. To setup, set the `url` for your book and the path to your `cover-image` file. Your book's `title` and `description` are also used.

This `gitbook` uses the same social sharing data across all chapters in your book— all links shared will look the same.

Specify your book's source repository on GitHub using the `edit` key under the configuration options in the `_output.yml` file, which allows users to suggest an edit by linking to a chapter's source file.

Read more about the features of this output format here:

<https://pkgs.rstudio.com/bookdown/reference/gitbook.html>

Or use:

```
?bookdown::gitbook
```

